



**DELHI PUBLIC SCHOOL NEWTOWN**  
**SESSION 2023-24**  
**MONDAY TEST**

**CLASS: X**  
**SUBJECT: BIOLOGY**

**FULL MARKS: 40**  
**DATE: 19.6.23**

**General Instructions:**

- The paper consists of two printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

1. Choose the correct answer: [5]
  - a. Karyokinesis is the division of:
    - i. Cytoplasm, ii. Nucleus, iii. Chromosome, iv. Nucleolus.
  - b. Main nitrogenous waste present in urine is:
    - i. Uric acid, ii. Creatinine, iii. Urea, iv. Amino acids.
  - c. Ultrafiltration occurs in:
    - i. Bowman's Capsule, ii. Loop of Henle, iii. DCT, iv. PCT.
  - d. The Basal Metabolic Rate in the body cells is regulated by:
    - i. Parathyroid, ii. Adrenal gland, iii. Pancreas, iv. Thyroid.
  - e. The region where crossing over takes place is known as:
    - i. Spindle fibre, ii. Centromere, iii. Chiasmata, iv. Synapsis.
2. Choose the odd one out and write the correct category to which the others belong: [5]
  - a. Mitosis, Karyokinesis, Cytokinesis, Chromosome.
  - b. Afferent arteriole, Efferent arteriole, Glomerulus, Pelvis.
  - c. Kidney, Ureter, Urethra, Uterus.
  - d. Chromatid, Chromosome, Chromatin thread, Centriole.
  - e. Goitre, Cretinism, Scurvy, Myxodema.
3. State the exact location of the following: [5]
  - a. Centromere
  - b. Pituitary gland
  - c. Thyroid gland
  - d. Kidney
  - e. Urinary bladder

4. Match the correct answers of Column I with Column II. Please rewrite the correct pairs: [5]

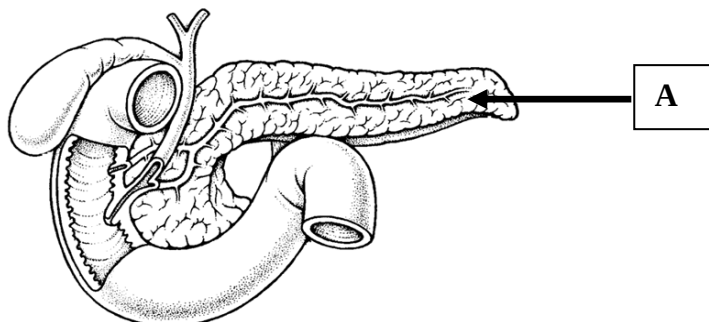
| Column I            | Column II                        |
|---------------------|----------------------------------|
| i. Bowman's Capsule | Pigment present in urine         |
| ii. Meiosis         | Blood in urine                   |
| iii. Haematuria     | Cortex of kidney                 |
| iv. ADH             | Chromosome number becomes half   |
| v. Urochrome        | Reabsorption of water            |
|                     | Anterior lobe of pituitary gland |
|                     | Glucose in urine                 |

5. Answer the following questions:

- Define nephron. [1]
- Differentiate between Histone and Nucleosome on the basis of their definition. [2]
- Why do we urinate more in winters than in summers? [2]
- Mention the two important features of Synthesis phase. [2]
- Draw a neat and well labelled diagram of late Anaphase. [3]

6. Answer the following questions:

- Define Tropic hormone. [1]
- Mention two important features of Metaphase. [2]
- What causes Ultrafiltration to take place in Glomerulus? [2]
- Name the gland from which we get cortisone. What do you understand by Adrenal virilism? [2]
- Observe the diagram given below and answer the following questions: [3]



- Identify the organ 'A'.
- Name the group of cells which are known as hormone secreting cells of "A".
- Mention the function of the hormone which is secreted from Beta cells of the above marked organ 'A'.